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Pyrolysis study of iso-propylbenzene with photoionization and molecular beam mass spectrometry

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ABSTRACT

Pyrolysis study of iso-propylbenzene (IPB) at 30 and 760 torr was carried out by using synchrotron photoionization and molecular beam mass spectrometry techniques. 0.5%IPB was diluted by helium and heated to 798–1148 K in a flow tube. Styrene was the most abundant aromatic intermediate at both pressures. Several polycyclic aromatics were only observed at 760 torr, such as 2-ethenyl-1, 4-dimethyl-benzene, fluorene, anthracene, phenanthrene, 2-methyl-anthracene and pyrene, while fulvene, 1, 3-cyclohexadiene, and 5-ethenylidene-1, 3-cyclopentadiene were only detected at 30 torr. A detail mechanism involving 306 species and 1990 reactions was developed by updating ipso-addition and unimolecular dissociation reactions of IPB as well as reactions related to styrene, indene and naphthalene. The mechanism could predict the pyrolysis of IPB and production of other intermediates well. Rate-of-production analysis shows that the formations of 2-phenethyl, 1-iso-phenylpropyl and 2-iso-phenylpropyl radicals are the main consumption reactions of IPB independent of pressure. Sensitivity analysis indicates that the unimolecular dissociation of IPB forming 2-phenethyl radical is the most significant promoting reaction at both pressures. H-abstraction reaction forming 1-iso-phenylpropyl radical tends to play a significant inhibiting effect on IPB consumption at 30 torr, while the most inhibiting reaction is self-combination of methyl radicals at 760 torr. Compared to n-propylbenzene experiments reported by Liu et al. (Combustion and Flame, 191 (2018) 53–65) and Yuan et al. (Combustion and Flame,186 (2017) 178–192.), IPB tends to produce larger quantities of benzene, styrene and indene under pyrolysis and oxidation conditions. These results will improve the understanding of the combustion and soot formation of IPB as a potential aviation surrogate fuel.

1. Introduction

The burning of fossil fuels is the main way for human beings to obtain energy, which also produces a large number of pollutants. Because of the complexity of the actual fuel composition, such as Jet-A and JP-8 [1,2], it is difficult to establish a detailed kinetic model and model fuels consisting of specific components are widely accepted to emulate the actual fuel. Alkyl aromatic hydrocarbons are widely found in gasoline, diesel oil and aviation kerosene, which have significant impacts on the formation of soot [3–9]. The addition of C_9 alkyl aromatic hydrocarbons would improve characteristics of surrogate mixtures such as molecular weight and distillation curve for the target Jet-A fuel [10]. Specifically, iso-propylbenzene (IPB) and n-propylbenzene (NPB) are usually used as model fuels for their complex branched chain structures which could represent the branched alkylbenzene contents of actual fuels [11,12]. Therefore, the studies of propylbenzenes (NPB and IPB) are very important to further understand the formation pathways of soot precursors, improve the combustion technology of fuel and reduce pollutant emissions in the combustion of aviation kerosene.

Previous studies on propylbenzenes mainly concentrated on the combustion characteristics by means of shock tube, laminar and counter-flow diffusion flame [13–16]. The laminar flame propagation velocity of NPB was measured and simulated by Hui et al. [13]. Rauband et al. [14] studied the ignition delay time of NPB in the range of 500–600 K at 24.7 atm. Wang et al. [15] reported the species profiles in low-pressure premixed NPB flame under the fuel-rich condition by using synchrotron photoion-

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ization and molecular beam mass spectrometry techniques and detected a large number of free radicals. Conturso et al. [16] compared the particle formation tendency among NPB, IPB and 1,3,5-trimethylbenzene in atmospheric counter-flow diffusion flames. Liu et al. [11] studied the NPB combustion characteristics by conducting low temperature oxidation experiments in a jet-stirred reactor (JSR), measuring laminar flame speeds in laminar flames and ignition delay times in a shock tube. Quite few works have been reported on the detailed reaction kinetics of IPB [12,17]. Litzinger et al. [17] conducted high temperature oxidation experiment of IPB in a flow tube under atmospheric pressure from 900 to 1300 K, and assumed that H-abstraction, ipso–addition and unimolecular decomposition reaction were three main pathways of IPB consumption. Besides, Wang et al. [12] studied the low temperature oxidation of IPB in a JSR within the temperature range of 700–1100 K at the pressure of 1 atm and equivalence ratios of 0.4 and 2.0, and developed a detailed oxidation model involving 306 species and 1985 reactions. As the initial reaction of combustion, pyrolysis reaction is the main way of fuel decomposition and production of small molecule products. Meanwhile, pyrolysis experiments under different pressures could provide a clear understanding about the initial decomposition pathways of IPB as a model component of aviation fuels. However, studies on the pyrolysis of IPB were rarely reported.

This work focused on the pyrolysis characteristics of IPB in a flow tube over the temperature range of 798–1148 K at the pressures of 30 and 760 torr with synchrotron radiation photoionization and molecular beam mass spectrometry (MBMS) techniques. Based on the pyrolysis results and the work of Wang et al. [12], a kinetic model was updated. The Rate-of-production (ROP) and sensitivity analysis were conducted to reveal the reaction routes of IPB and formation channels of main pyrolysis intermediates which could help to improve the understanding of soot precursor formation in IPB pyrolysis process.

2. Experimental methods

The pyrolysis experiments were carried out at the National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The detailed description of the instrument setup has been reported elsewhere [18–21], and only a brief description is given here. The experimental apparatus consists of four parts: a pyrolysis chamber equipped with an electric heating furnace and a flow tube, a differential pumped chamber with a molecular-beam sampling system, a photoionization chamber and a home-made reflection time of flight mass spectrometer (RTOF-MS). The material of the flow tube is made of α-alumina which could inhibit wall catalytic effects [22], and the heating section of the α-alumina tube is 220 mm long and the inner diameter was 7 mm.

The temperature profiles in the furnace alone the axis of the flow tube under different pressures were measured by an S-type thermocouple. The uncertainty of the measured temperature profiles is estimated to be within $\pm$30 K [23]. The measured temperature profiles are shown in Fig. S1 in the supplementary materials (SM). IPB was purchased from Aladdin Reagent (Shanghai) Co., Ltd., Shanghai, China, with a purity of $\geq$99.5%. IPB was injected into the evaporator by a chromatographic pump and vaporized at 453 K, then was carried out by He as the carrier gas. The gas components flowing into the pyrolysis furnace are IPB (0.5%), helium (98.5%) and krypton (1%) and the total flow rate was 1000 sccm. The pressures in the pyrolysis chamber were kept at 30 and 760 torr, respectively, which were controlled by a MKS throttle valve. The reactants and products were sampled by a quartz nozzle at the tip and crossed a nickel skimmer to form supersonic molecular beam which were ionized by synchrotron VUV light and detected by the RTOF-MS [19,24]. The wavelength range of the ultraviolet light is 0.19–0.08 µm. Because the initial total flow rate is constant, the residence time is affected by pressure and temperature changes. The calculated residence times at 30 and 760 torr in the studied temperature range are 5.2–6.2 $\times$ 10^{-3} and 1.3–1.6 $\times$ 10^{-1} s.Two modes were adopted in the IPB pyrolysis experiments. One mode was to measure a series of mass spectra with the variation of photon energy at specified temperatures to identify the chemical structures of the pyrolysis products. In this study, the temperatures were 1073 K at 30 torr and 948 K at 760 torr. The other mode was to record mass spectra at the fixed photon energies with the variation of temperature to yield mole fractions versus temperature. In the present work, the measurements of temperature scan were carried out at several selected photon energies of 13.00, 11.00, 10.50, 10.00, 9.50 and 9.00 eV. The temperature ranges were from 948 to 1148 K at 30 torr and from 798 to 1073 K at 760 torr. The analyzing method to obtain the species mole fractions has been introduced elsewhere [21]. For the species with known photoionization cross section data [25] (see Table S1 in the supplemental materials, SM), the obtained concentration is within 25% uncertainty. For the species with unknown photoionization cross section data, the uncertainty of concentration is a factor of two.

3. Kinetic model

A detailed kinetic model of IPB pyrolysis which consisted of 306 species and 1990 reactions was updated based upon our recent work with respect to IPB oxidation [12]. The thermochemical data were also referred to the IPB oxidation mechanism [12] without modification. The simulated results using IPB oxidation mechanism [12] are shown in Figs. S2 and S3 (see SM). It is clear that the IPB oxidation mechanism [12] underestimated the onset decomposition temperatures of IPB at both pressures. Meanwhile, the formation temperatures of C_2H_2, C_2H_4 and styrene ($A1C_2H_3$) and the mole fractions of propene (C_3H_6), α-methylstyrene ($A1C(CH_3)CH_2$) and indene (C_9H_8) were underestimated. Moreover, the mole fractions of naphthalene (A2), anthracene (A3a), and phenanthrene (A3p) were overestimated.

In the present model, the unimolecular dissociation reactions of IPB forming 1-iso-phenylpropyl ($A1CH(CH_3)CH_2$), 2-isophenylpropyl ($A1C(CH_3)_2$) and 2-phenylethyl ($A1CHCH_3$) radicals were referred to those of ethylbenzene ($A1C_2H_5$) [26–29]. The rate constant of $A1CH(CH_3)_2(+M)$=A1-+$iC_3H_7(+M)$ considering the pressure effect was referred to $A1CH_3(+M)$=A1-+$CH_3(+M)$ [30]. The decomposition of $A1C(CH_3)_2$ forming $A1C(CH_3)CH_2$ was referred to the reaction of $A1CHCH_3$=$A1C_2H_3$+H [29,31]. The ipso-addition reaction of IPB by H atom producing benzene (A1) and iC_3H_7 was estimated from the reaction $A1C_2H_5$+H = A1+C_2H_5 [32], which was larger than that in the oxidation model [12]. Such reactions make iC_3H_7 radicals formation faster than in the oxidation model [12].

The unimolecular dissociation of $A1C_2H_5$ forming benzyl radical ($A1CH_2$) was updated by referring to [26–29]. The formation of $A1C_2H_3$ from 1-phenylethyl ($A1CH_2CH_2$) and $A1CHCH_3$ radicals via β–scission were referred to [29,31]. Formation reactions of C_9H_8 from $A1CH_2$ combining with C_2H_2 and $A1C_2H$ combining with CH_3 were referred to [33]. The rate constants of $A1CCH_2$+C_2H_2=A2+H were estimated according to the reaction C_2H_2+iC_4H_5=H + A1 [34]. Formation reaction of A3p via phenyl radical (A1-) combining with $A1C_2H$ was referred to C_4H_4+C_2H_3=A1+H [35]. The rates of A2 and A3p formation reactions were significant lower than those in the oxidation model [12]. The comparison of reaction rate coefficients is shown in Fig. S4 (see SM). The newly added and modified reactions in IPB mechanism is shown in Table. S5 (see SM).

The pyrolysis results were simulated using PFR mode of CHEMKIN software [36]. Sensitivity and ROP analysis were carried out based on the modeling results. Moreover, the present model

Table 1
Intermediates and products identified in the pyrolysis of iso-propylbenzene.

m/z	Formula	Species	IE/eV			30 torr			760 torr		
			Ref	30 torr	760 torr	X_M	T_F/K	T_M/K	X_M	T_F/K	T_M/K
26	C_2H_2	Acetylene	11.4	–	–	5.72E–05	1073	1148	1.85E–04	948	1073
28	C_2H_4	Ethylene	10.5	–	–	3.49E–04	1023	1148	6.41E–04	873	1073
30	C_2H_6	Ethane	11.52	–	–	1.85E–04	998	1148	1.69E–04	873	1023
40	C_3H_4	Allene	9.69	9.75	9.7	4.21E–05	1023	1148	1.00E–04	898	1073
42	C_3H_6	Propene	9.73	9.75	9.75	1.60E–04	973	1123	2.42E–04	823	973
44	C_3H_8	Propane	10.94	–	–	–	–	–	1.62E–06	898	998
52	C_4H_4	Vinylacetylene	9.58	9.55	9.55	1.09E–05	998	1148	7.85E–06	873	1073
54	C_4H_6	1,3-Butadiene	9.07	9.05	9.05	1.39E–05	998	1148	4.92E–05	898	1023
56	C_4H_8	2-Butene	9.1	9.15	9.1	3.13E–05	1023	1123	4.29E–05	873	998
66	C_5H_6	1,3-Cyclopentadiene	8.57	8.55	8.55	2.53E–05	1023	1148	1.56E–05	898	1073
68	C_5H_8	1,3-Pentadiene	8.59	–	8.55	–	–	–	5.33E–06	873	998
78	C_6H_6	Fulvene	8.36	8.4	–	4.21E–06	1023	1123	–	–	
		Benzene	9.24	9.25	9.25	3.42E–04	998	1148	8.86E–04	823	1023
80	C_6H_8	1,3-Cyclohexadiene	8.25	8.2	–	1.12E–05	1023	1148	–	–	
90	C_7H_6	5-Ethenylidene-1,3-Cyclopentadiene	8.29	8.25	–	4.31E–06	1073	1148	–	–	
91	C_7H_7	*Benzyl Radical*	7.24	7.25	–	2.40E-06	948	1148	–	–	
92	C_7H_8	Toluene	8.79	8.8	8.8	5.02E-04	1048	1148	1.97E-04	873	1073
102	C_8H_6	Phenylethyne	8.82	8.8	8.8	1.75E-04	1073	1148	2.29E-04	948	1073
104	C_8H_8	Styrene	8.47	8.45	8.45	3.95E-03	998	1148	2.08E-03	823	998
106	C_8H_{10}	Ethylbenzene	8.77	8.75	8.75	1.71E-04	998	1148	1.11E-04	823	998
116	C_9H_8	Indene	8.15	8.15	8.15	7.12E-05	1048	1148	8.25E-05	898	1073
118	C_9H_{10}	α-Methylstyrene	8.3	8.3	8.25	9.53E-04	973	1123	1.00E-03	798	973
128	$C_{10}H_8$	Naphthalene	8.15	8.15	8.15	4.60E-06	1048	1148	3.07E-05	948	1073
132	$C_{10}H_{12}$	2-Ethenyl-1,4-Dimethyl- Benzene	8	–	8	–	–	–	3.65E-05	873	973
166	$C_{13}H_{10}$	Fluorene	7.88	–	7.9	–	–	–	3.66E-06	973	1073
168	$C_{13}H_{12}$	4-Methyl-1,1′-Biphenyl	7.8	7.75	–	3.68E-06	1073	1148	–	–	
178	$C_{14}H_{10}$	Anthracene/Phenanthrene	7.47/7.86	–	7.5/7.85	–	–	–	3.15E-05	948	1023
180	$C_{14}H_{12}$	Stilbene/cis-Stilbene	7.66/7.8	7.65	7.65/7.8	2.16E-05	1048	1148	3.66E-05	998	1023
192	$C_{15}H_{12}$	2-Methyl-Anthracene	7.37	–	7.4	–	–	–	5.77E-06	948	1023
202	$C_{16}H_{10}$	Pyrene	7.45	–	7.45	–	–	–	1.79E-06	998	1073

Note: Ref means data are obtained by consulting literature.
X_M refers to the maximum molar fraction in the experimental temperature range.
T_F refers to initial temperature for formation of species.
T_M refers to the temperature relating to the peak mole fraction;.
Radical has been emphasized in italics.

was further validated against the measured data obtained under oxidation conditions.

4. Results and discussion

4.1. Species identification

Based on the measurement of photoionization mass spectra and the photoionization efficiency spectra (PIE), about 30 species were identified at 30 and 760 torr, as shown in Table 1. Photoionization data of species are referenced from NIST database [37]. Several PAHs such as fluorene ($C_{13}H_{10}$), A3a, A3p and pyrene (A4) and small species such as C_2H_2, vinylacetylene (C_4H_4) and 1,3-butadiene (C_4H_6) were not observed in oxidation experiment of IPB [12]. It could be attributed to that such unsaturated hydrocarbons are important precursors to form PAHs and could be consumed quickly under oxidation conditions. Nevertheless, some species were only observed at low pressure such as 1,3-cyclohexadiene (C_6H_8) compared to 760 torr. PAHs such as $C_{13}H_{10}$, A3a, A3p and A4 could only be found at 760 torr, and their PIE spectra are shown in Fig. 1. This difference is attributed to higher probability of molecular collision at 760 torr.

4.2. Mole fraction profiles

Figures 2-4 show the mole fraction profiles of IPB and its pyrolysis products mole fractions against temperature. In general, the simulation results could reproduce the experimental well by present mechanism. With the increase of pyrolysis pressure, IPB begins to decompose at lower temperature. C_8 species tend to form under lower pressure. However, polycyclic aromatics tend to be generated more abundantly at higher pressure. Besides, small molecules are definitely formed at both pressures, but in larger concentrations at higher pressure.

The experimental and simulated profiles of IPB and monocyclic aromatics are shown in Fig. 2. The present model predicts the IPB consumption well, although it tends to slightly overestimate IPB consumption at 760 torr. IPB begins to decompose at 998 K under the pressure of 30 torr. This value shifts to 848 K under the pressure of 760 torr, which indicated that the IPB begins to decompose at a lower temperature under higher pressure. It should be noted that IPB begins to consume at 775 K under fuel-lean condition [12], significantly lower than that under pyrolysis condition. It can be attributed to that H-abstractions by H radicals and unimolecular dissociation are the dominant consumption pathways for IPB pyrolysis at both pressures which need higher temperature to trigger.

$A1C_2H_3$ is found as the most abundant aromatic product with maximum mole fractions of 3.95×10^{-3} at 30 torr and 2.08×10^{-3} at 760 torr in the IPB pyrolysis process. At low pressure, IPB pyrolysis forms more $A1C_2H_3$ than at atmosphere where $A1C_2H_3$ tended to convert to PAHs such as stilbene ($C_{14}H_{12}$). In the pyrolysis process, IPB undergoing unimolecular decomposition and H-atom abstraction, is more favorable to break down into $A1CHCH_3$ and $A1CH(CH_3)CH_2$ radicals and then directly to form $A1C_2H_3$ by β–scission.

A1 and $A1C(CH_3)CH_2$ are also major aromatic compounds. A1 is mainly formed via the ipso-addition of IPB by H atom at both pressures. The maximum mole fraction of A1 is 3.42×10^{-4} at 30 torr, and this value reach 8.86×10^{-4} at 760 torr. IPB pyrolysis at low pressure tend to form less A1. The peak values of $A1C(CH_3)CH_2$

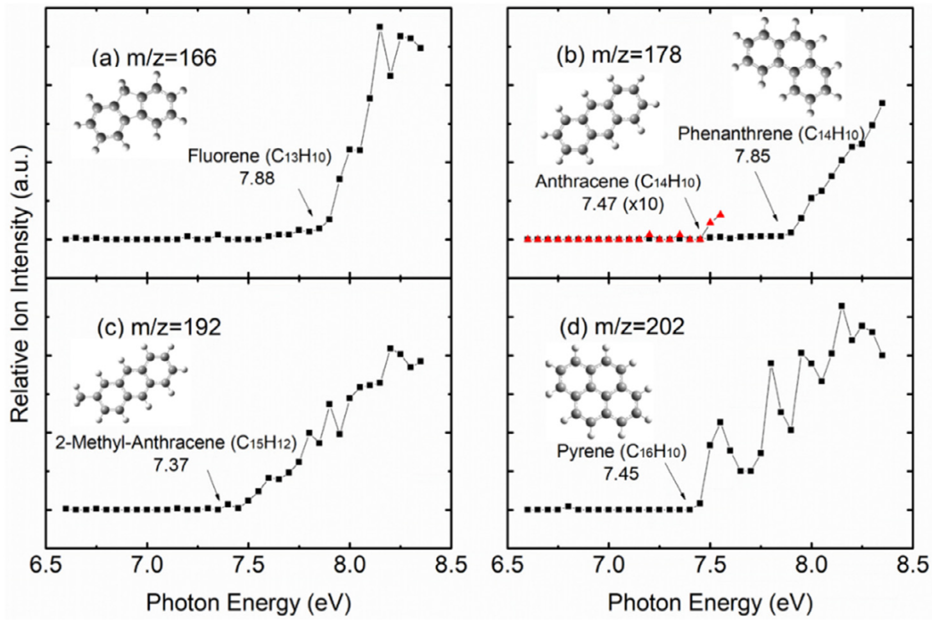

Fig. 1. PIE spectra obtained in the pyrolysis of iso-propylbenzene at the pressure of 760 torr. Numbers in the panels represent ionization energy (IE) of species.

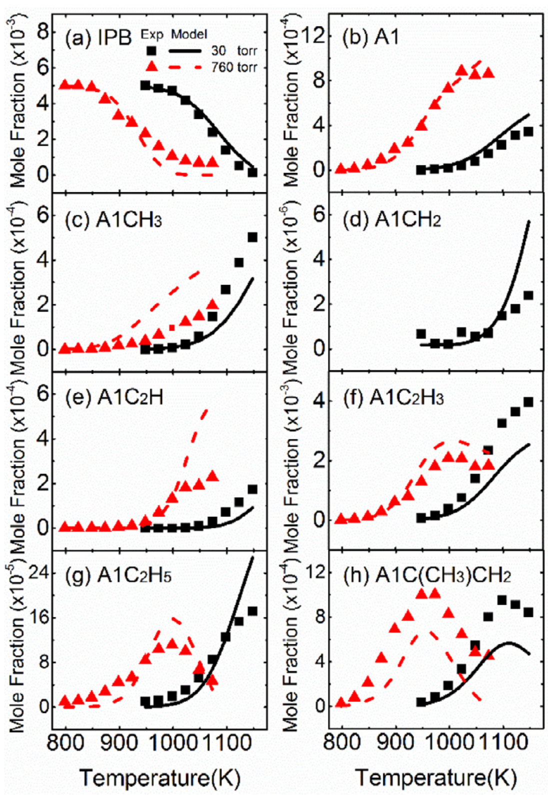

Fig. 2. Experimental (symbols) and simulated (lines) mole fraction profiles of iso-propylbenzene (IPB), benzene (A1), toluene ($A1CH_3$), benzyl radical ($A1CH_2$), phenylacetylene ($A1C_2H$), styrene ($A1C_2H_3$), ethylbenzene ($A1C_2H_5$), and α-methylstyrene ($A1C(CH_3)CH_2$) in the pyrolysis of IPB at 30 and 760 torr.

formation are independent with the pressures, through the temperature against the peak value drop by 150 K with the pressure increasing from 30 to 760 torr.

$A1CH_2$ was only significantly observed at the pressure of 30 torr with the maximum mole fraction of 2.40×10^{-6}. The model reproduces satisfactorily the mole fraction profile of $A1CH_2$ radicals.

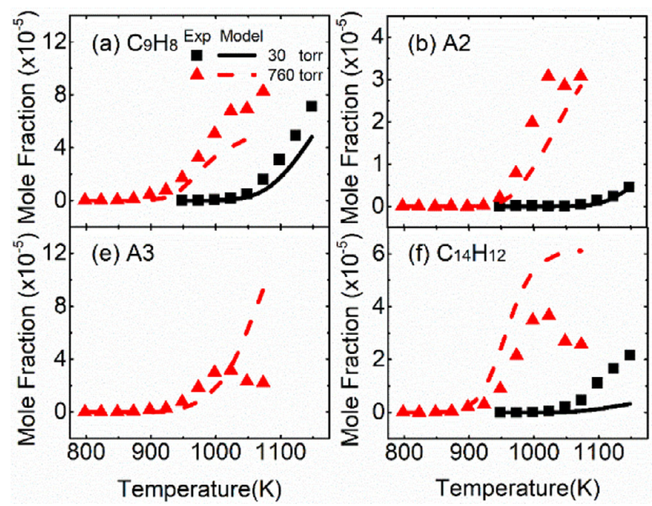

Fig. 3. Experimental (symbols) and simulated (lines) profiles of PAH species in the pyrolysis of IPB at 30 and 760 torr.

$A1CH_2$ radicals are mainly formed from IPB dissociation at both pressures, which will then combine with H, CH_3 and A1- radicals to give $A1CH_3$, $A1C_2H_5$ and diphenylmethane ($C_{13}H_{12}$). At higher pressure, there is a higher probability of annihilation of $A1CH_2$ by colliding with other molecules or radicals which would cause its concentration becoming too low to be detected. Besides, $A1CH_2$ was not detected in the oxidation conditions [12] due to the limitation of GC to detect free radicals.

Figure 3 presents the comparison between experimental data and simulated results for PAH species. The pressure has a strong effect on the onset temperature and the yield for PAHs. The yield of naphthalene increased significantly at higher pressure. The yield of $C_{14}H_{12}$ is about two times at 760 torr compared 30 torr. A3 (A3a and A3p) formation was not observed at 30 torr, through it formed obverously at 760 torr. While there is no significant change on the formation of C_9H_8. A3 is overestimated by a factor of 3 and other species are modeled with a factor better than 2. Further research on coupling reactions among A3 and other aromatics need to be

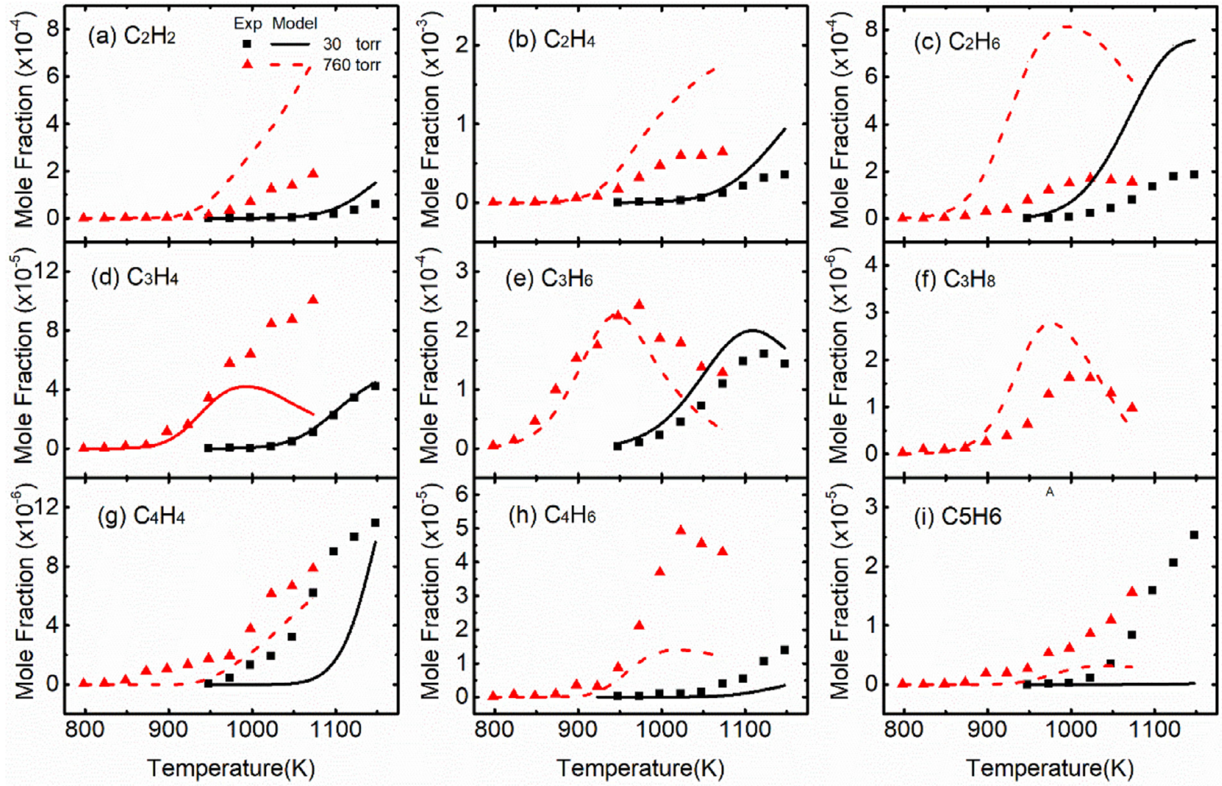

Fig. 4. Experimental (symbols) and simulated (lines) profiles of C_2-C_5 species in the pyrolysis of IPB at 30 and 760 torr.

considered in the present mechanism which could improve the accuracy of prediction about A3.

Figure 4 displays the comparison for the profiles of C_2–C_5 species. The formations of C_2H_2, C_2H_4, allene (aC_3H_4), C_3H_6 and C_4H_6 increase significantly with the pressure increasing apart from C_2H_6, C_4H_4 and C_5H_6. C_2 species were overestimated while C_4H_6 and C_5H_6 were underestimated. According to ROP analysis, CH_3 radicals which are formed mainly from the unimolecular dissociation reaction of the fuel, are important precursors to generate C_2 species. Most of CH_3 radicals convert to C_2H_6 by self-combined reaction. The formation of C_2 species are overestimated obviously. It could be attributed to that the consumption pathways of CH_3 radicals and C_2 species combining with PAHs are not complete. aC_3H_4 is mainly formed from A1C(CH_3)CH_2 decomposition. C_3H_6 is mainly generated from iC_3H_7 radicals via β–scission. iC_3H_7 radicals generated mainly from ipso-addition of IPB with H and CH_3 radicals are important precursors of C_3H_6. Propane (C_3H_8) is only observed at the pressure of 760 torr with the maximum mole fraction of 1.62×10^{-6}, which is mainly formed from C_2H_5 radical combining with CH_3 radical. With respect to C_4 species, most of them are mainly formed from C_3 species reacting with CH_3 radicals. For example, C_4H_4 is mainly formed from C_3H_2 radicals reacting with CH_3 radicals at 30 torr. For C_4H_6 and C_5H_6, they are products of multi-step decomposition of IPB, which are affected by many elementary reactions. Some elementary reactions related to its formation or consumption may need to be improved.

In general, C_2H_2, aC_3H_4, C_3H_8, C_4H_4 and C_4H_6 were not observed at oxidation conditions [12]. C_2H_2, aC_3H_4, and C_3H_6 were only observed at 30 and 150 torr in the pyrolysis experiments of NPB, which were detected at 30 and 760 torr in the pyrolysis experiments of IPB. The reason had been introduced in [38]. Besides the formation pathways of C_2–C_3 species in IPB pyrolysis are very different from in NPB pyrolysis [38].

4.3. Sensitivity and ROP analysis

Sensitivity analysis and ROP analysis for pyrolysis of IPB at 30 torr (52% conversion at 1073 K) and 760 torr (53.4% conversion at 948 K) are displayed in Figs. 5 and 6. As shown in Fig. 5, unimolecular decomposition at the side isopropyl group forming A1CHCH$_3$ radical exhibits the most significant promoting effect at both pressures. Abstraction of H atom from the side isopropyl group generating A1CH(CH_3)CH_2 radical tends to play a significant inhibiting effect on IPB consumption at 30 torr, while it plays a promoting effect at 760 torr. It could be attributed to that, part of A1CH(CH_3)CH_2 radicals isomerize directly to A1CH_2CHCH$_3$ radicals at 30 torr. However, most of A1CH(CH_3)CH_2 radicals decomposed to A1C_2H_3 and CH_3 radicals via β-scission, and CH_3 radicals play important roles in IPB consumption at 760 torr. The most inhibiting reaction changed to CH_3+CH_3(+M)=C_2H_6(+M) at 760 torr. However, At lean condition, H-abstraction by OH radical at the side isopropyl group of IPB forming A1CH(CH_3)CH_2 radical has the most significant promoting effect [12]. H and CH_3 radicals play important roles in the pyrolysis of IPB. For the oxidation of IPB, OH radicals play more significant role than CH_3 radicals, while CH_3 radicals tend to play more important roles under fuel-rich condition [12].

The ROP analysis is shown in Fig. 6. There are three main consumption pathways for IPB pyrolysis, namely unimolecular decomposition forming A1CHCH$_3$ radical, and H-abstraction by H and CH_3 forming A1CH(CH_3)CH_2 and A1C(CH_3)$_2$ radicals, respectively. The proportions of these three consumption pathways are very close, and all are around 30%, regardless of pressure. H abstractions of IPB by H atom forming A1CH(CH_3)CH_2 and A1C(CH_3)$_2$ radicals was account for 55.61% at 30 torr and 56.52% at 760 torr. The proportions of H abstraction of IPB by CH_3 are very small, only 1.28% at 30 torr and 4.78% at 760 torr. In the oxidation experiments [12], IPB mainly converts to A1CH(CH_3)CH_2 radical at lean

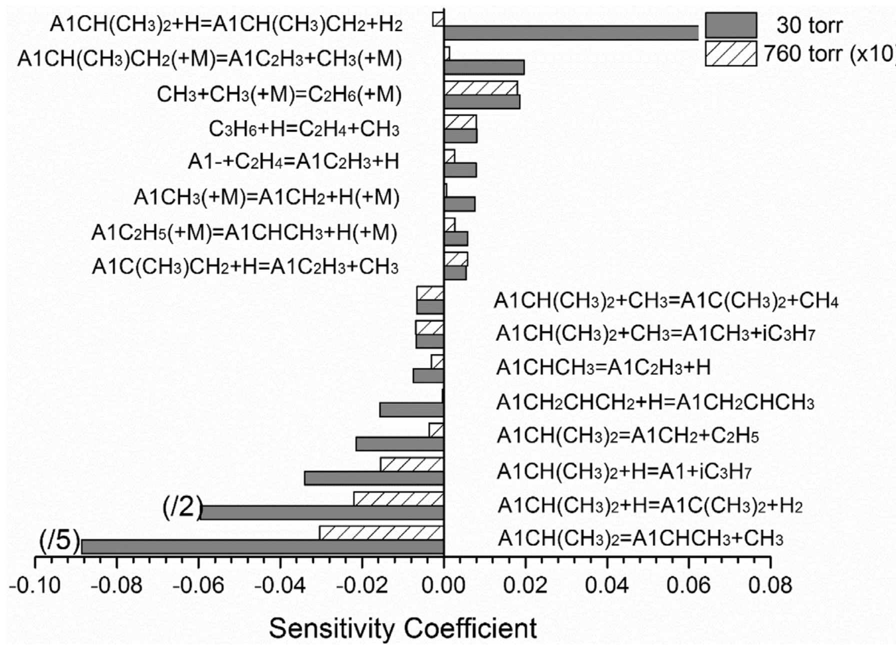

Fig. 5. Sensitivity analysis of pyrolysis of IPB ($A1CH(CH_3)CH_2$) at a conversion ratio of about 55%. The pressures are 30 (1073 K) and 760 torr (948 K).

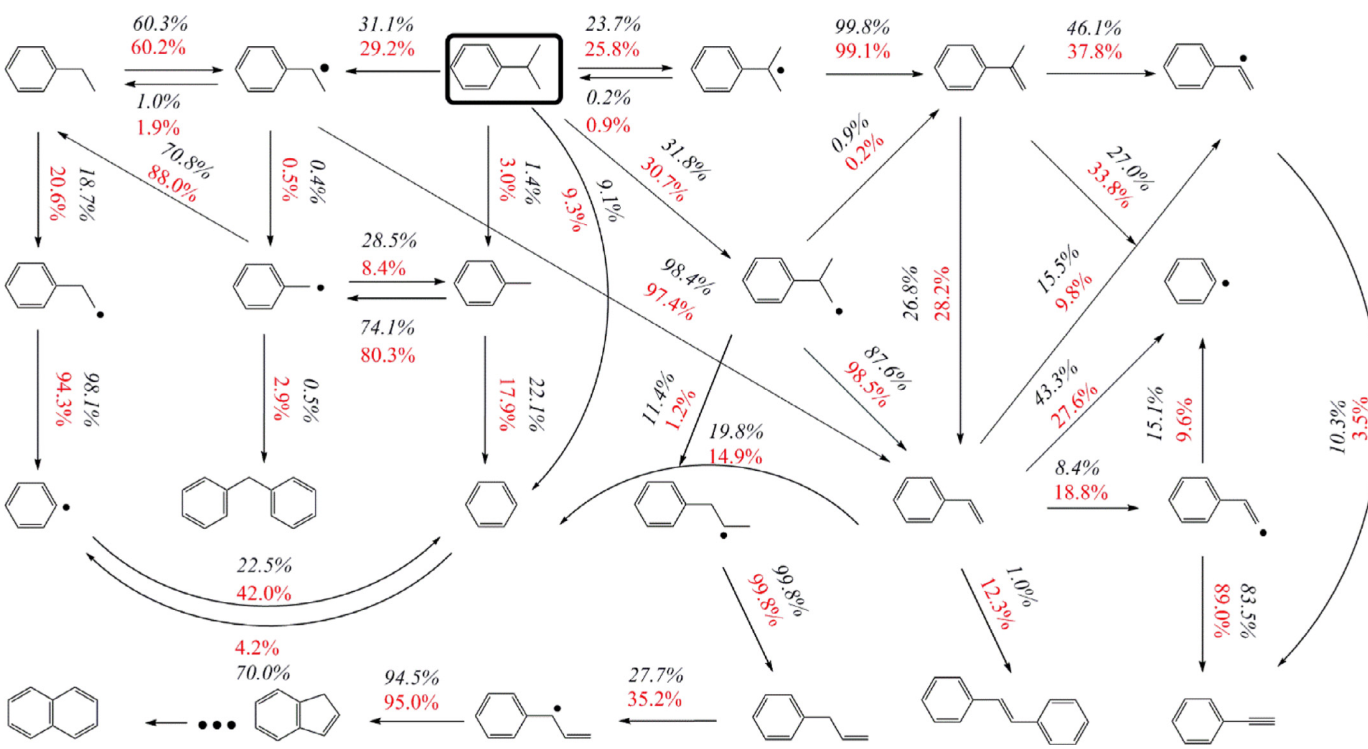

Fig. 6. ROP analysis of pyrolysis of IPB shown for a conversion ratio of about 55%: italic numbers–$P = 30$ torr at $T = 1073$ K; normal numbers – $P = 760$ torr at $T = 948$ K.

(79%) and rich (68%) conditions. The H-abstractions of IPB forming $A1CH(CH_3)CH_2$ and $A1C(CH_3)_2$ radicals are the dominant consumption channels for IPB oxidation. OH radicals play the most important role in the H abstractions of IPB, which is different from the pyrolysis condition.

β–scissions of C–H and C–C bonds play leading roles in the formation process of $A1C_2H_3$ and $A1C(CH_3)CH_2$. Most of $A1CHCH_3$ (98.47% at 30 torr and 97.45% at 760 torr) and $A1CH(CH_3)CH_2$ radicals (87.67% at 30 torr and 98.55% at 760 torr) decompose to form $A1C_2H_3$ via β–scission. $A1C(CH_3)CH_2$ is the dominant product of $A1C(CH_3)_2$ radicals (99.80% at 30 torr and 99.10% at 760 torr) via β–scission. Nevertheless, this formation pathway was not considered in the mechanism of Wang et al. [12]. $A1C(CH_3)CH_2$ is consumed mainly by H-atom attacking and unimolecular dissociation reaction to give A1- radicals, $A1CCH_2$ radicals and $A1C_2H_3$. Nearly 25% of $A1C(CH_3)CH_2$ converts to $A1C_2H_3$, which was not observed in oxidation conditions [12].

As for the most abundant aromatic product, $A1C_2H_3$ is a precursor of many aromatic species, such as A1, phenylacetylene ($A1C_2H$) and stilbene ($C_{14}H_{12}$). $A1C_2H_3$ is mainly consumed by the H-addition, giving rise to A1- and C_2H_4. Other important consumption pathways of $A1C_2H_3$ are the ipso-addition reaction by H

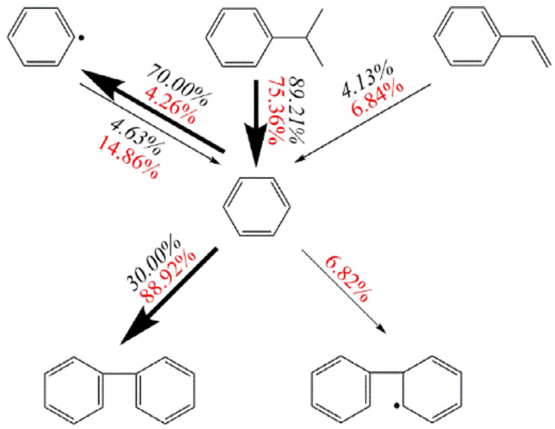

Fig. 7. ROP analysis for the formation and consumption of A1 under pyrolysis conditions, for a conversion ratio of about 55% of IPB by using the present model: italic numbers–$P = 30$ torr at $T = 1073$ K; normal numbers – $P = 760$ torr at $T = 948$ K.

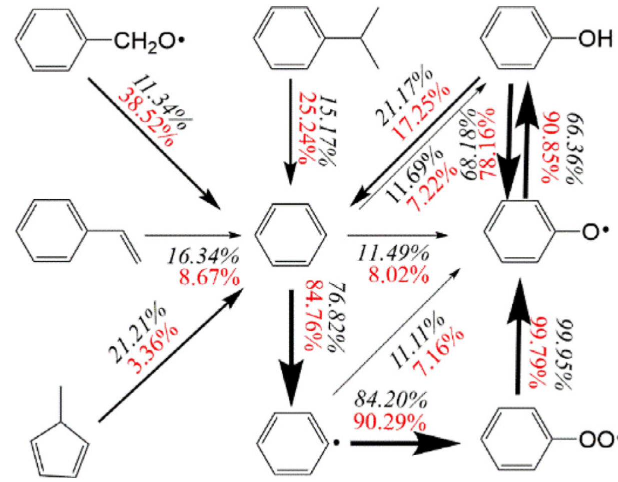

Fig. 8. ROP analysis for the formation and consumption of A1 under oxidation conditions, for a conversion ratio of about 60% of IPB by using the oxidation model [12]: italic numbers – $\phi = 2.0$ at $T = 905$ K; normal numbers – $\phi = 0.4$ at $T = 850$ K.

atom producing A1 and the vinyl radicals, and the H-abstraction reactions leading to 1-phenylvinyl (A1CHCH) and 2-phenylvinyl ($A1CCH_2$). $A1C_2H_3$ could also combine with A1- radical to produce $C_{14}H_{12}$, and the proportion of this reaction varies significantly at different pressures. The proportion at 760 torr is 12.30% and higher than 1.07% at 30 torr. Due to the lack of oxygen, the pathway of $A1C_2H_3$ to A1CHCHO radicals observed in the oxidation process [12] is not appeared in this work.

$A1C_2H$ is mainly formed from the decomposition of A1CHCH and $A1CCH_2$ radicals. Most of A1CHCH radicals decompose to $A1C_2H$ at both pressures, and only a small fraction of A1CHCH radicals decompose to A1-. The pressure has no significant effect on the decomposition of A1CHCH radicals. However, the consumption pathway of $A1CCH_2$ radicals is greatly affected by pressure. At 30 torr, 10.31% of $A1CCH_2$ radicals decompose to $A1C_2H$, while this proportion drops to 3.52% at 760 torr.

$A1C_2H_5$, toluene ($A1CH_3$) and A1 are also important intermediates during the pyrolysis of IPB. $A1C_2H_5$ is mainly formed from $A1CH_2$ radical combining with CH_3 radical, which is quite similar to the IPB oxidation at rich condition [12]. $A1CH_3$ is mainly formed by ipso-addition via CH_3 radical at 760 torr. However, it is mainly formed from the reaction of $A1CH_3$ (+M)=$A1CH_2$+H (+M) at 30 torr. Ipso-addition by H atom on IPB molecule could directly produce A1, which accounts for the major pathway of A1 production at both pressures (89.21% at 30 torr and 75.36% at 760 torr).

The formation and consumption pathways of A1 under pyrolysis and oxidation conditions are shown in Figs. 7 and 8. Figure 7 displays the main formation and consumption pathways of A1 at pyrolysis conditions. It could be noted that A1- radicals, IPB and $A1C_2H_3$ are important precursor of A1. A1 tend to form A1- radicals by H-abstractions at 30 torr and more likely combine with A1- radicals to form biphenyl (P2) at 760 torr. Besides, A1- radicals could combine with CH_3 radicals and form $A1CH_3$ at 30 and 760 torr. Figure 8 shows the formation and consumption pathways of A1 at oxidation conditions. Benzyl hydroxyl ($A1CH_2O$) is an important precursor of A1 which will not exit at pyrolysis conditions. Same as the pyrolysis conditions at 30 torr, H-abstractions forming A1- radicals is the main consumption pathway at lean and rich conditions.

Besides the mono-aromatic species, C_9H_8 and A2 were also produced abundantly in the pyrolysis process. The leading formation pathways of the C_9H_8 and A2 are shown in Fig. 9. In general, pressure has great influence on the formation routes of C_9H_8 and A2. 85.54% of C_9H_8 is formed via the reaction of C_9H_8+H = $A1CHCHCH_2$ at 30 torr. This proportion decreases dramatically to 20.83% at 760 torr. At 30 torr, the for-

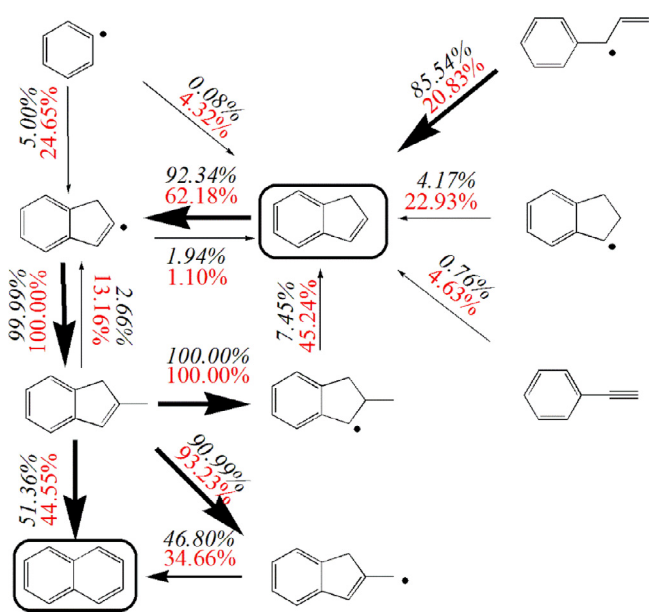

Fig. 9. Formation pathways of Indene (C_9H_8) and naphthalene (A2) of IPB pyrolysis: italic numbers–$P = 30$ torr at $T = 1073$ K; normal numbers – $P = 760$ torr at $T = 948$ K. The number represents the percent of products generated from its precursor.

mation of A2 followed the reaction sequence: C_9H_8 → indenyl radical (C_9H_7) → methylindene ($C_9H_7CH_3$) → indenemethyl radical ($C_9H_7CH_2$) → A2. $C_9H_7CH_3$ could also convert to A2 directly. C_9H_8 is mainly consumed via H-abstractions by H atom and CH_3 radicals to give the resonance stabilized C_9H_7 radicals. Most of C_9H_7 radicals combine with CH_3 radicals and formed $C_9H_7CH_3$. $C_9H_7CH_2$ radical is mainly formed by H-abstraction and then converted to A2. At 760 torr, nearly half of C_9H_8 is generated from the reaction of C_9H_8+CH_3(+M)=$C_9H_8CH_3$(+M), which proportion is only 7.45% at 30 torr. A2 is mainly produced from the reactions of CH_3+$C_9H_7CH_3$=CH_4+A2+H and H + $C_9H_7CH_3$=H_2+A2+H at both pressures. In the NPB pyrolysis process, C_9H_8 is mainly formed by cyclization and dehydrogenation reactions of 1-phenyl-propen-2-yl ($A1CHCCH_3$) and 1-phenyl-propen-3-yl ($A1CHCHCH_2$) radicals [38]. A2 could be produced from $A1CH_2$ radical by propargyl radical addition and cyclization reactions or be produced from $A1C_2H_3$ H-abstraction, acetylene addition and cyclization reactions [38].

Table 2
Experimental conditions for oxidation and pyrolysis of IPB and NPB.

Exp. conditions	NPB		IPB	
	Oxidation	Pyrolysis	Oxidation	Pyrolysis
Pressure (torr)	760		760	
Temperature range (K)	700–1100	800–1200	700–1100	798–1073
Reactor	JSR	Flow Tube	JSR	Flow Tube
Equivalence ratio	0.4 and 2.0	$+\infty$	0.4 and 2.0	$+\infty$

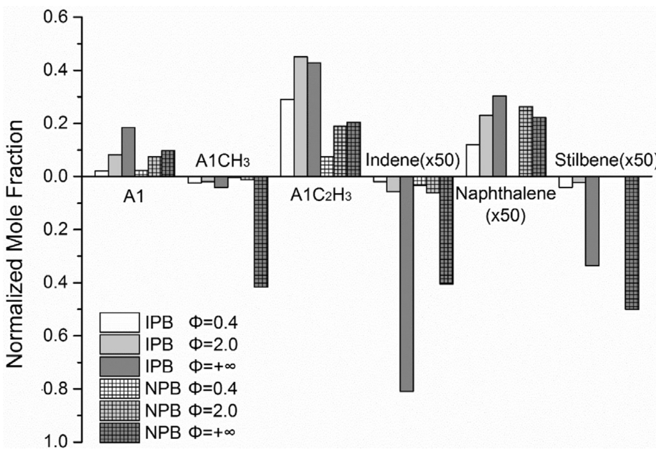

Fig. 10. Comparison of representative intermediates in low-temperature oxidation and pyrolysis of NPB and IPB [11,12,38].

4.4. Comparison between IPB and NPB under oxidation and pyrolysis conditions

Although IPB and NPB have the same chemical formula, the molecular structures of IPB and NPB are significantly different, which obviously affect the formation of intermediate products. In addition, the equivalence ratio has a significant effect on the types and yields of radicals, which affects the consumption process of reactants. To find the effect of fuel structure and equivalence ratio on the production of representative intermediates, comparison was performed among the two C_9H_{12} alkylbenzenes [11,12,38], as presented in Fig. 10. The experimental conditions of C_9H_{12} species oxidation and pyrolysis [11,12,38] were shown in Table 2. In order to eliminate the effect of different import concentration of fuel, the concentration of all intermediates was normalized (divided by the fuel inlet concentration) and defined as normalized mole fraction. A1, $A1CH_3$ and $A1C_2H_3$ are the most abundant mono aromatics both in the low-temperature oxidation and pyrolysis of IPB and NPB. However, NPB pyrolysis tends to produce more $A1CH_3$. Since the benzylic C–C bond (BDE = 71.7 kcal/mol) of NPB is the weakest [39], and its dissociation could produce $A1CH_2$ which was an important precursor to form $A1CH_3$. The formation of $A1CH_3$ in NPB pyrolysis was higher than in IPB pyrolysis. However, the formation of $A1C_2H_3$ is lower than in IPB pyrolysis, which has been mentioned above. In general, NPB pyrolysis tends to form $A1CH_2$ radicals which are the main precursors to form $A1CH_3$ by adding H-atom. However, IPB tends to form $A1CH(CH_3)CH_2$, $A1C(CH_3)_2$ and $A1CHCH_3$ radicals which could produce $A1C_2H_3$ by β-scission. As the equivalence ratio increases, the formation of A1 increased significantly. There is more A1 formed in the pyrolysis of IPB compared to the pyrolysis of NPB. However, there is no significant difference on A1 formation at oxidation conditions between IPB and NPB. The formation pathways of A1 between IPB oxidation and pyrolysis are quite different. In oxidation conditions, A1 is mainly formed via the reaction of A1OH+H=A1+OH and $CH_3+C_5H_5CH_3$=>CH_4+A1+H, while $A1CH(CH_3)_2$+H=A1+iC_3H_7 is the main source of A1 formation in pyrolysis conditions.

C_9H_8, A2 and $C_{14}H_{12}$ were all detected in the oxidation and pyrolysis of both IPB and NPB. There is no significant difference in the formation of C_9H_8 between IPB and NPB oxidation. IPB pyrolysis tends to produce more indene than NPB pyrolysis. A2 was not detected in the lean condition of NPB low temperature oxidation, and its production reduced when the experiment condition changed to pyrolysis from rich condition of NPB low temperature oxidation. However, A2 formation in IPB combustion increased with the increase of equivalence ratio. Similar phenomena also occurred in the formation of $A1C_2H_3$ during the pyrolysis and oxidation of IPB. It could be attributed to that such two species are important precursors of many PAHs and tend to combine with other molecules or radicals to form PAHs at pyrolysis conditions, which would consume a lot of $A1C_2H_3$ and A2 molecules. Besides, $C_{14}H_{12}$ was not identified in the low temperature oxidation of NPB, which was observed with a significant production in pyrolysis. In general, with the increase of equivalent ratio, the production of aromatics increased significantly.

4.5. Validation against oxidation data

In order to extend the equivalence ratios ranges of these simulations and verify the correctness of the updated present mechanism of IPB, the results of Wang et al. [12] have also been modeled. Figure. S5 (see SM) presents the comparison of simulation results against oxidation data of IPB at lean condition between the present mechanism and the mechanism of Wang et al. [12]. The present mechanism reproduces satisfactorily the consumption of IPB and the formation of the main intermediates and products, through the prediction of IPB consumption is slightly slower than the measurements and predictions by the Wang's model [12].

The ROP analysis of IPB oxidation are shown in Fig. S6 (see SM). While the consumption paths of IPB are very similar to those obtained by using the oxidation model [12], several differences appear in the consumption paths between these two models. IPB is mainly consumed by H-abstractions to give the resonance stabilized $A1CH(CH_3)CH_2$ and $A1C(CH_3)_2$ radicals both in the oxidation model and the present model. Nevertheless, proportion of unimolecular dissociation of IPB forming $A1CHCH_3$ radicals decreased to 0% compared to 2% in the oxidation model. The difference in the formation of $A1CHCH_3$ radicals is related to the difference in reaction mechanisms. The updated rate constant of unimolecular dissociation of IPB forming $A1CHCH_3$ radicals is slower than before which could weaken the $A1CHCH_3$ formation. Besides, the reaction of $A1C(CH_3)_2$ radicals converting to $A1C(CH_3)CH_2$ via β–scission was completely negligible in the oxidation model, which is a major formation path of $A1C(CH_3)CH_2$ in the present model.

In general, there were no significant differences on the flux analysis of IPB oxidation between the Wang's model [7] and this work. It could be attributed to that the present mechanism was mainly updated the pressure-related unimolecular dissociation reactions, although unimolecular dissociation reaction had little effect on fuel consumption under fuel-lean condition.

5. Conclusions

This work presents new experimental and modeling results for IPB pyrolysis in a flow tube at 30 and 760 torr. With the pressure increasing, the onset temperature of IPB pyrolysis decreases significantly from 998 K at 30 torr to 848 K at 760 torr. Besides, the formation of PAHs increases with increasing pressure. The simulated results display reasonable agreement with the measured results. Unimolecular dissociation at the side isopropyl group forming $A1CHCH_3$ radical is the most significant promoting reaction for IPB pyrolysis. H abstraction reaction at the side isopropyl group forming $A1CH(CH_3)CH_2$ radical is the most inhibiting reaction at 30 torr, while the most inhibiting reaction change to $CH_3+CH_3(+M)=C_2H_6(+M)$ at 760 torr. Unimolecular dissociation and H-abstraction of IPB molecule forming $A1CH(CH_3)CH_2$, $A1C(CH_3)_2$ and $A1CHCH_3$ radicals are the main consumption pathways of IPB pyrolysis. The model has also been validated against the experimental results of IPB oxidation with a good agreement. Moreover, the comparison of experimental results between IPB and NPB with a wide range of equivalence ratios was made. IPB pyrolysis tends to produce more PAHs compared to low temperature oxidation; IPB pyrolysis and oxidation tend to produce more A1, $A1C_2H_3$ and C_9H_8 compared to NPB pyrolysis and oxidation. These results will be beneficial for future work on understanding the combustion and soot formation of IPB and other alkylbenzenes comprehensively.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.07.036.

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